Examiner only

3. Some Year 10 students were given three unknown white solids – **A**, **B** and **C**.

They carried out a series of flame tests and silver nitrate tests to identify the solids.

Their results are shown in the table.

	Observations	
Solid	Flame test	Silver nitrate test
A	apple-green flame	cream precipitate
B	red flame	white precipitate
C	yellow flame	yellow precipitate

(a) Name solids **A**, **B** and **C**. [3]

A ....

B ....

C ....

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(b) Complete and balance the symbol equation for the reaction between magnesium chloride and silver nitrate. [2]

$MgCl_2 + \boxed{} AgNO_3 \longrightarrow$ ....

(c) 0.103 g of silver nitrate, $AgNO_3$, was used to make up a solution.

Calculate the number of moles of silver nitrate in this mass. Give your answer in **standard form**. [3]

$A_r(Ag) = 108$ $A_r(N) = 14$ $A_r(O) = 16$

Number of moles = ... mol

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